

Supplementary Material for RibbonTok: Multi-Resolution, Single-Stream, and Adaptive-Length Tokenization for Autoregressive Image Generation

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Extended Related Work

Discrete visual representations. VQ-VAE introduced end-to-end learning of discrete latent codes, and VQGAN combined perceptual/adversarial reconstruction with transformer modeling (van den Oord, Vinyals, and Kavukcuoglu 2017; Esser, Rombach, and Ommer 2021). Later work improves codebook learning, replaces vector lookup with finite scalar levels, or composes residual codes (Yu et al. 2022; Mentzer et al. 2024; Lee et al. 2022). Most such representations originate from a 2D grid. MaskGIT and Muse model grid tokens with masked prediction (Chang et al. 2022, 2023); MAGVIT-v2, Open-MAGVIT2, MaskBit, LlamaGen, and Emu3 strengthen discrete tokenization or large-scale token generation (Yu et al. 2024a; Luo et al. 2024; Weber et al. 2024; Sun et al. 2024; Wang et al. 2026). MAR instead uses continuous tokens and a per-token diffusion objective (Li et al. 2024). These systems establish strong reconstruction and generation baselines, but their token count and spatial organization are generally selected for a particular operating point.

Compact and flexible 1D tokenizers. TiTok resamples an image into a compact register sequence (Yu et al. 2024b). ALIT uses recurrent allocation, ElasticTok learns nested elastic representations, CAT controls content-dependent capacity, and One-D-Piece couples causal prefixes with externally estimated complexity (Duggal et al. 2025; Yan et al. 2025; Shen et al. 2025; Miwa et al. 2025). FlexTok is especially close in its ordered registers, nested dropout, and flow decoder (Bachmann et al. 2025). ReToK studies information over-concentration, AdaTok adds a learned content-conditioned budget policy, and ChannelTok orders channel slices to obtain a lightweight flexible representation (Fu et al. 2026; Lu et al. 2026; Paul, Bansal, and Goldstein 2026). Concurrent CaTok binds causal intervals to a MeanFlow decoder (Chen et al. 2026). RibbonTok retains an externally selected budget and instead focuses on a factorized discrete position plus cross-resolution assignment alignment.

Resolution-general versus resolution-consistent representations. VibeToken directly decouples dynamic 1D to-

ken length from input and output raster size, supports broad resolutions and aspect ratios, and is therefore the closest resolution-general comparison (Patel et al. 2026). RibbonTok does not claim a broader rendering range. Its additional target is assignment-level consistency: views with the same logical detail but different physical rasters receive a shared common-prefix assignment distribution during training. This distinction matters because a decoder may accept many target grids even when independently resized inputs map to unrelated token identities. Conversely, assignment alignment does not by itself establish arbitrary-resolution generalization. The two properties are complementary; we use “ID” only for an explicit objective that aligns discrete common-prefix assignments across such views.

Generative detokenization. Latent diffusion and diffusion transformers demonstrate high-capacity generation in compressed spaces (Rombach et al. 2022; Peebles and Xie 2023). Rectified flow and flow matching learn velocity fields along probability paths (Liu, Gong, and Liu 2023; Lipman et al. 2023), with scalable transformer recipes improving high-resolution synthesis (Esser et al. 2024). DiTo further treats diffusion-based reconstruction as a tokenizer objective (Chen et al. 2025). Such detokenizers are useful when compact codes are ambiguous, but they can synthesize details not uniquely determined by those codes. This motivates our terminology of conditional reconstruction and our separate reporting of paired metrics, component latency, and full generation latency.

Detailed Method

Causal Register Encoding and Directional Quantization

Let the patchified VAE latents be $\mathbf{z}_{1:N}$ and the learned registers be $\mathbf{r}_{1:L}$. In every ViT block, latent queries attend only to latent keys. Register query ℓ attends to all latent keys and register keys $1:\ell$. Thus $\mathbf{u}_\ell$ may summarize the complete image but cannot depend on future visual positions.

For codebook k , let $\mathbf{u}_\ell^{[k]}$ be the corresponding chunk and $\mathbf{e}_j^{[k]}$ be the unit code vector for entry j . The cosine-assignment logits are

$$s_{\ell j}^{[k]} = \left\langle \frac{\mathbf{u}_\ell^{[k]}}{\|\mathbf{u}_\ell^{[k]}\|_2}, \mathbf{e}_j^{[k]} \right\rangle, \quad \|\mathbf{e}_j^{[k]}\|_2 = 1. \quad (1)$$

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At evaluation, $y_\ell^{[k]} = \arg \max_j s_{\ell j}^{[k]}$. The complete visual position is $\mathbf{y}_\ell = (y_\ell^{[1]}, \dots, y_\ell^{[K]})$. The tuple space has cardinality V^K , but this combinatorial capacity does not imply uniform occupancy of all tuples.

Multi-Resolution Assignment and Codebook Optimization

For view $i \in \{0, 1, 2\}$, position ℓ , and codebook k , temperature-scaled logits define

$$p_{\ell j}^{(i)[k]} = \frac{\exp(s_{\ell j}^{(i)[k]}/\tau)}{\sum_{m=1}^V \exp(s_{\ell m}^{(i)[k]}/\tau)}. \quad (2)$$

Before the softmax, coalesce matching replaces the logits at every common-prefix position by

$$\bar{s}_{\ell j}^{[k]} = \frac{1}{3} \sum_{i=0}^2 s_{\ell j}^{(i)[k]}, \quad 1 \leq \ell \leq L^{(1)} = L^{(2)}. \quad (3)$$

The full-resolution view retains its own logits for $L^{(1)} < \ell \leq L^{(0)}$. Training samples

$$y_\ell^{(i)[k]} \sim \text{Categorical}(p_{\ell 1:V}^{(i)[k]}), \quad (4)$$

and passes the selected code vector with a straight-through estimator,

$$\tilde{\mathbf{u}}_\ell^{(i)[k]} = \mathbf{u}_\ell^{(i)[k]} + \text{sg} \left(\mathbf{e}_{y_\ell^{(i)[k]}}^{[k]} - \mathbf{u}_\ell^{(i)[k]} \right), \quad (5)$$

where sg denotes stop-gradient. Code vectors are updated by an exponential moving average of assigned chunks. The encoder receives

$$\mathcal{L}_{\text{commit}} = \sum_{i,k,\ell} \left\| \mathbf{u}_\ell^{(i)[k]} - \text{sg} \left(\mathbf{e}_{y_\ell^{(i)[k]}}^{[k]} \right) \right\|_2^2. \quad (6)$$

Let m index all assignment distributions for codebook k in one minibatch. We combine low token-wise entropy with high aggregate entropy,

$$\mathcal{L}_{\text{entropy}} = \sum_{k=1}^K \left[\frac{1}{M} \sum_{m=1}^M H(\mathbf{p}_m^{[k]}) - H \left(\frac{1}{M} \sum_{m=1}^M \mathbf{p}_m^{[k]} \right) \right], \quad (7)$$

where $H(\mathbf{p}) = -\sum_j p_j \log p_j$. The first term favors decisive assignments; the negative aggregate term favors broad marginal code use.

Scope of the consistency claim. Coalescing makes the three common-prefix assignment distributions identical during a triplet training pass. Independent categorical samples may still differ, providing stochastic regularization, whereas deterministic argmax under the coalesced logits gives identical tuples. At ordinary single-view inference, no companion views are required: each image is tokenized independently, so cross-view identity is a learned property rather than an algebraic invariance guarantee. By contrast, rendering one already obtained prefix on any supported target grid follows directly from the cross-attention interface. We use *resolution-consistent* for the trained assignment behavior and report its finite evaluated scope explicitly.

Prefix Flow Matching

For each view, we independently sample a retained length $q^{(i)} \sim \mathcal{U}\{1, \dots, L^{(i)}\}$ and give only $\tilde{\mathbf{u}}_{1:q^{(i)}}^{(i)}$ to the detokenizer. For clean VAE latents $\mathbf{z}_0^{(i)}$, Gaussian noise ϵ , and $t \sim \mathcal{U}(0, 1)$, the interpolation and velocity objective are

$$\mathbf{z}_t^{(i)} = (1-t)\mathbf{z}_0^{(i)} + t\epsilon, \quad (8)$$

$$\mathcal{L}_{\text{fm}}^{(i)} = \left\| \mathbf{v}_\theta \left(\mathbf{z}_t^{(i)}, t, \tilde{\mathbf{u}}_{1:q^{(i)}}^{(i)} \right) - (\epsilon - \mathbf{z}_0^{(i)}) \right\|_2^2. \quad (9)$$

All trainable tokenizer and detokenizer modules are optimized by

$$\mathcal{L} = \frac{1}{3} \sum_{i=0}^2 \mathcal{L}_{\text{fm}}^{(i)} + \lambda_c \mathcal{L}_{\text{commit}} + \lambda_e \mathcal{L}_{\text{entropy}}. \quad (10)$$

The VAE is fixed. At inference, assignments are deterministic and the detokenizer integrates from Gaussian noise on the requested supported latent grid.

Factorized Autoregressive Interface

Each visual position is embedded by four independent lookup tables whose outputs are concatenated into the transformer width. Four output heads produce the next-position logits. With class or null condition $\mathbf{c}$, the loss is

$$\mathcal{L}_{\text{AR}} = - \sum_{\ell=1}^L \sum_{k=1}^K \log p \left(y_\ell^{[k]} \mid \mathbf{c}, \mathbf{y}_{1:\ell-1} \right). \quad (11)$$

After top- p truncation, the four indices at one position are sampled in parallel. A generated stream may terminate at any externally selected length represented during training.

Training and Implementation Details

Tokenizer. The fixed VAE has 16 latent channels and spatial downsampling factor 8. We sample from its posterior during training and use its posterior mode at evaluation, followed by a fixed latent scaling factor. The latent grid is patchified with patch size 2. All variants use $L_{\text{max}} = 256$ registers, token dimension $d = 256$, $K = 4$ directional codebooks, and $V = 16,384$ entries per codebook. Linear projections reconcile the ViT, token, and DiT widths.

Images are resized and randomly cropped at the maximum target resolution, with random horizontal flipping. The lower logical resolution is sampled uniformly from the training schedule; area downsampling produces $\mathbf{x}^{(1)}$, and nearest-neighbor interpolation produces $\mathbf{x}^{(2)}$. Training uses 20K optimizer steps and a global batch size of 3,840. AdamW uses $\beta = (0.9, 0.99)$, weight decay 10^{-3} , peak learning rate 3.84×10^{-4} , 1K linear warmup steps, cosine decay, and gradient clipping at 3. Assignment temperature decreases linearly from 0.2 to 0.05 over the first 4K steps. EMA decay is 0.995, $\lambda_c = 0.25$, $\lambda_e = 0.05$, and straight-through residual dropout is 0.2.

Metric		M	L	XL
Params. (M)	ViT	85	287	287
	DiT	127	578	2170
	Total	296	949	2541
FLOPs (T)	Tok.	0.36	0.56	0.56
	Detok.	4.99	15.34	53.67
Memory (GB)	Base	1.32	3.58	9.51
	Peak	1.52	3.79	9.73
Latency (ms)	Tok.	16.2	20.9	21.2
	Detok.	827.1	1206.5	2192.4

Table 1: Single-image RibbonTok component profile on one A100. All variants process 256×256 images with the full 256-position prefix; detokenization uses 25 rectified-flow steps. “Tok.” is VAE-latent tokenization through discrete assignment, and “Detok.” is latent-grid generation through pixel decoding. The 2,192.4-ms XL detokenizer is the 2.19-s component used in the end-to-end comparison; AR latency is measured separately from these tokenizer and detokenizer components.

Autoregressive models. The 1B and 3B generators follow Llama-style causal-transformer configurations (Meta AI 2024). Class, null-condition, and beginning-of-sequence embeddings are learned separately; conditioning is dropped with probability 0.2. Both models are trained without image augmentation using a global batch size of 1,280. Training has three 10K-step stages at maximum sequence lengths 64, 128, and 256. In each stage, AdamW uses $\beta = (0.9, 0.95)$, weight decay 10^{-2} , peak learning rate 1.128×10^{-3} , 500 warmup steps, cosine decay, and gradient clipping at 1. Headline results use 32 positions, top- $p = 0.95$, CFG 3.0, and 25 detokenization steps.

Codebook Capacity and Utilization

Figure 1 shows that increasing vocabulary size generally improves reconstruction until gains diminish around 16K entries. Multiple codebooks outperform a single codebook at comparable per-codebook size, but aggressive factorization becomes counterproductive when chunks are too small or assignments become sparse. The default $K = 4, V = 16,384$ balances these effects.

For hard validation assignments, let $p_j^{[k]}$ be the empirical frequency of entry j in codebook k . We report normalized perplexity

$$\text{NP} = \frac{1}{K} \sum_{k=1}^K \frac{\exp\left(-\sum_{j=1}^V p_j^{[k]} \log p_j^{[k]}\right)}{V}, \quad (12)$$

which lies in $[1/V, 1]$. Figure 2 gives an NP of 0.97 for the default. This indicates broad marginal use within each codebook; it does not establish uniform use of all code tuples. Larger vocabularies or more chunks eventually reduce utilization and coincide with diminishing or negative reconstruction returns.

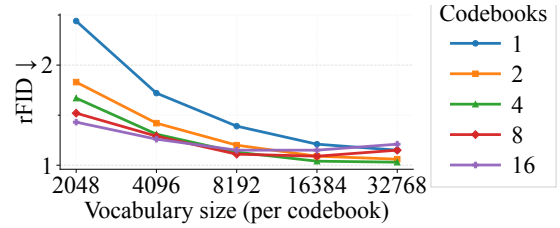


Figure 1: Directional-codebook capacity versus reconstruction. ImageNet-val-256 rFID for RibbonTok-XL with the full 256-position prefix and 25 detokenization steps. The sweep varies the number and size of directional codebooks while keeping the remaining tokenizer/detokenizer setup fixed; gains diminish near the default four 16K-entry codebooks.

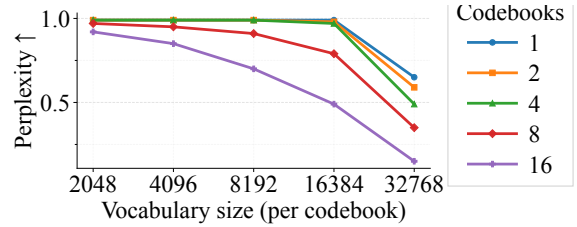


Figure 2: Codebook utilization for the same capacity sweep. Normalized perplexity averages the empirical hard-assignment perplexity of each codebook and divides by its vocabulary size. The default $K = 4, V = 16,384$ setting reaches 0.97, indicating broad marginal usage but not uniform occupancy of the V^K tuple space.

Additional Qualitative Results and Limitations

The current system has four main limitations. First, the token budget is externally selected rather than inferred from image content; in the title and text, adaptive length refers to one model supporting multiple selected lengths. Second, resolution consistency is trained and evaluated on a finite family of square targets, whereas VibeToken demonstrates a broader range of resolutions and aspect ratios. Third, the present experiments do not yet report a direct independent-view token-identity score; native-grid reconstruction and the coalesce ablation are system-level evidence. Fourth, small text and sharp glyphs remain difficult, and the 25-step flow detokenizer is the largest fixed cost at short budgets. Future work should learn calibrated stopping, add direct cross-view assignment diagnostics, extend training across aspect ratios and higher resolutions, and accelerate flow integration.

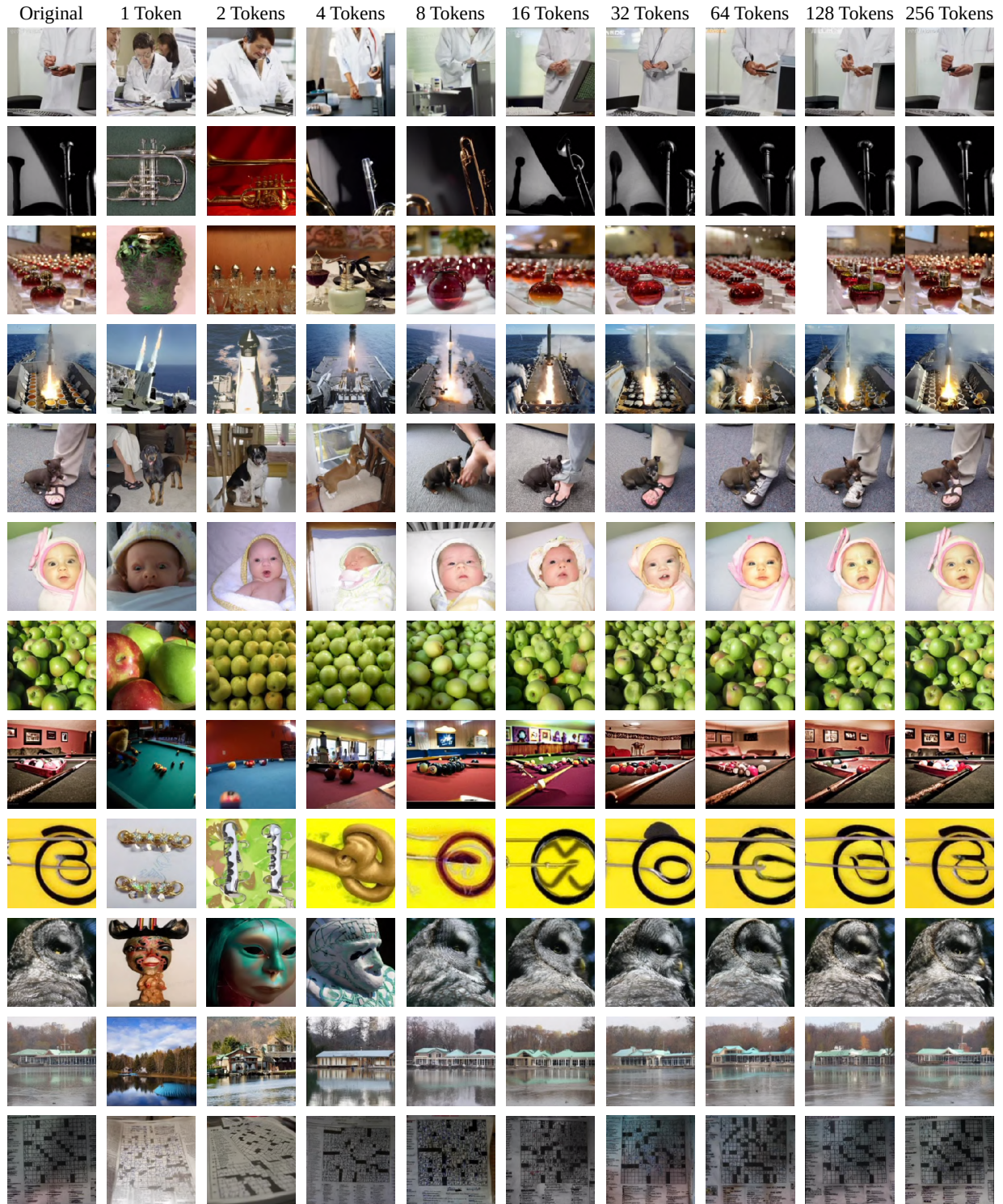


Figure 3: Additional nested-prefix reconstructions on ImageNet. Each row starts from one validation image and decodes progressively longer prefixes with the same RibbonTok-XL detokenizer and target grid. Early positions preserve class, layout, and dominant appearance; later positions increasingly constrain texture and instance-specific detail. Variation under short prefixes should be interpreted as conditional completion by the generative detokenizer, not deterministic recovery of unencoded pixels.

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